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Chapter 2

BARNSTORMING IN SPACE: THE RISE AND FALL
OF THE ROMANTIC ERA OF SPACEFLIGHT, 1957-1986

Alex Roland

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THUCYDIDES
The Peloponnesian War

Across the seamless web of history, historians stitch seams. Not to distort the record, or to mislead the reader, but to make sense out of chaos. History is a never-ending Bayeux tapestry, interweaving lives and deaths that overlap one another and shape one another. Any story lifted from it must begin and end somewhere. Choosing the limits of a historical tale is part of the historian's responsibility. It is a way of organizing knowledge, making it comprehensible and discrete.

The eras historians define become the benchmarks of our historical understanding, the frames in which we come to see and thus comprehend our past. Renaissance humanists, for example, gave us the notion of the Dark Ages because they saw themselves as standing on a cultural plateau comparable to that of Rome at its height. Between their achievement and Rome's they viewed a cultural chasm into which Western society had fallen for a millennium. The term dark ages was exaggerated and misleading, but it has nonetheless shaped our historical consciousness for five

hundred years. The seams between historical periods may be dramatic and instantaneous, or they may be subtle and invisible. The French Revolution, for example, separates the early modern era from the modern era with a violent and blood-red stitch. The transition from the classical world to the middle ages was stretched out over the slow and painful decline and fall of the Roman Empire. Historians choose their thread and their stitch to match the circumstances they envision.

The most difficult seams to discern are those that haven't been invented yet. If the nuclear age began in 1945, when did it or will it end? When did we pass into the post-industrial age? When did the electronic age begin? Is the Cold War over? We can guess at these eras, but only time will confirm which stitches hold, which are rent. So too with attempts to define the space age. In the wake of the Apollo 11 mission, Arthur Schlesinger, Jr., said that when historians look back on the twentieth century they will recall it above all else as the time when man began the exploration of space. This is more cautious than Richard Nixon's hyperbolic assertion that the eight days of that mission marked "the greatest week in the history of mankind since the Creation," but it is hardly the kind of detached judgment one might expect of a historian.¹ If the space age endures as a useful historical concept, it will need to be developed with more perspective than Professor Schlesinger was able to bring to it in the first flush of the Apollo achievement. Surely we still view the space age as a useful historical concept, but it is not as clear as it seemed in 1969 that the moon landing was the defining event.

I propose that we are now between the first and second stages of the space age. The first stage clearly began with Sputnik, and, as Professor Schlesinger hypothesized, it was focused on man's initial departure from planet earth. The end, however, was not the moon landing but the Challenger accident. From 1957 until 1986, the world went through what I call the romantic era of spaceflight, characterized by a succession of dramatic manned space spectacles in earth orbit and cislunar space. The nature of the second period is as yet unclear, but it is likely to be much different. Equally unclear is whether Challenger will mark a dramatic transition comparable to the French Revolution or whether the romantic era of spaceflight will pass slowly and perhaps painfully into some more advanced stage. In either event, the transition is under way.

A romantic first stage in human developments is not an unfamiliar phenomenon, especially when the developments surround technological revolutions. Sputnik's ability to escape

the surface of the earth shocked the human imagination; when Yuri Gagarin followed in 1961, the effect was even more profound. Through technological ingenuity, modern man had separated his generation from all others that went before, setting off flights of imagination of the kind that transported Professor Schlesinger. These events created what the historian Carl Becker called a new "climate of opinion." Seventeenth century in origin, the term refers to those "instinctively held preconceptions in the broad sense, that *Weltanschauung* or world pattern."² When fostered by technological revolutions like Sputnik or manned spaceflight, new climates of opinion often take the form of romantic enthusiasms for a utopian future that seems ordained by the new technology, a kind of manifest destiny of the machine that seems just as sure and just as important to the initiate as religious belief is to the faithful.

Similar enthusiasm greeted the introduction of steam power in the nineteenth century, making of James Watt a national and international hero and transforming the age's view of its future. Electricity had a similar effect when it was finally reduced to human service at the hands of another international hero, Thomas Edison. Henry Adams captured some of this wonderment surrounding these two achievements when he compared the "silent and infinite force" of the dynamos at the Great Exposition of 1900 with the moral force that "early Christians felt [for] the Cross" and that medieval pilgrims and peasants felt for the Virgin Mary in her cathedral at Chartres.³ The dynamo and the virgin, he came to believe, were symbols and drivers of their respective ages. Closer to our own time, nuclear power seemed to open up boundless opportunities for those who understood its potential, leading Atomic Energy Commission Chairman Lewis Strauss to predict that electricity would soon be "too cheap to meter."⁴ The "green revolution" of the following decade sparked equally utopian visions of a world without hunger.⁵ Indeed, utopianism has become one of the concomitants of technological breakthroughs. Nuclear power and the new agricultural technology have transformed our world forever, but not in the way the enthusiasts first envisioned. Star Wars and the National Aerospace Plane are just as likely to fall short of the hyperbolic rhetoric surrounding their introduction.⁶

The nearest parallel to the romantic era of spaceflight came in the barnstorming era of aviation.⁷ Sputnik may be equated with the first flight of the Wright brothers in 1903, though of course the latter received far less public exposure. The Russian and American manned flights of 1961 parallel in many ways the flights

of Wilbur Wright in Paris and Orville Wright in Washington in 1908, which Charles Gibbs-Smith came to call the *annus mirabilis* of aviation, the year in which all the world came to recognize the true nature and the implications of the Wrights' achievement.⁸ Of course the moon landing of 1969 bears comparison with Lindbergh's 1927 crossing of the Atlantic, events watched and heralded around the world and making international heroes out of the Lone Eagle on the one hand and Neil Armstrong and his colleagues on the other. Through most of these adventures, the flyers themselves were almost as important in the public imagination as the technologies that carried them aloft. Eddie Rickenbacher, Wiley Post, and Amelia Earhart come as quickly to mind as Yuri Gagarin, John Glenn, and Sally Ride.⁹

The parallels between early aviation and spaceflight transcend the public persona of the flyers. Both technologies had immediately apparent civilian and military potentials, but in neither case was it entirely clear how they would be exploited.¹⁰ On the military side reconnaissance was seen as the first and most important role. Aircraft were in fact designed and used primarily for that purpose in World War I, where it was discovered that air superiority, close air support, and even strategic bombing held out still greater potential. So too has the military use of space been given over primarily to reconnaissance so far, though communications has moved increasingly to the forefront and Star Wars looms on the horizon.

Civilian uses of aviation were slower to develop, mail delivery giving way to passenger travel and finally large-scale freight transportation only well after the early romantic era was over; in spite of valiant early attempts, aerial photography and weather forecasting played only peripheral roles. In space, communications and scientific research have dominated civilian exploitation, with weather and earth resources playing lesser roles. Civilian aviation required large public subsidies, provided directly to national airlines in most countries and indirectly through research support and the Department of Commerce in the United States. So too in space, where scientific research has been subsidized almost entirely by the federal government, and the one commercial success so far, the communications satellite, has proceeded through a revolutionary public/private venture, COMSAT. Other civilian activities in space have been paid for almost entirely at public expense.

The closest parallel between the romantic era of aviation and the romantic era of spaceflight came in what historian Joseph Corn has called "the winged gospel," the almost religious

enthusiasm for the new technology that flourished in the United States in the early twentieth century.¹¹ A similar enthusiasm for manned spaceflight, a climate of opinion, holds sway over many Americans today. Bufts in the 1920s fastened on aviation firsts, like the first plane to commute from home to office, and my personal favorite, the first cat to fly the English Channel.¹² These may be compared with our recent infatuation with the first man in space, the first extra-vehicular activity, the first woman in space, and the first golf shot on the moon. Talk of a "space age" and "space-age technology" is reminiscent of earlier visions of an "aerial age," a "true" or "real" air age, a "new epoch," and a "mighty era" that promised a "great new future of the world."¹³ It was confidently predicted in the 1920s that explorers would henceforth make their discoveries from the air, that real estate agents would show property from on high, that travelling salesmen would use their planes as mobile showrooms, and that hunters would chase birds in flight.¹⁴ Today we hear analogous prophecies of cheap manufacturing in space, resorts in Earth orbit, and mail delivery by rocket. The most familiar theme of the winged gospel was "a plane in every garage," comparable to the myth of a Space Shuttle that "everyman" could fly.¹⁵ William Randolph Hearst's brilliant and cynical Junior Birdmen of America compares sadly with today's spacecamps, which build a political constituency by promising wide-eyed children an impossible future.¹⁶

One characteristic of the aviation visionaries of the 1920s is that they based their prophecies on the history that they knew. Alexander Graham Bell predicted in 1909 that the "aerial motor car" was just around the corner, displaying not unfettered vision but the most parochial myopia.¹⁷ As is often true of such utopianism, which is always a climate of opinion unto itself, it was both an exaggeration and an underestimation of what came to pass. The early visionaries failed to anticipate the high-speed commercial airliners that have shrunk the world to a commuting ground for jet setters and businessmen. Instead they took their own world of the emerging automobile and superimposed the airplane upon it.

So too have the visionaries of the early space age imagined futures that are mere extrapolations of our experience with aviation. On the one hand they have been moved to wildly unrealistic predictions of the volume and facility of spaceflight, building the Apollo launch facilities, for example, to handle fifty Saturn launches a year¹⁸ and predicating Shuttle operations on a similar estimate of fifty launches annually.¹⁹ As in the barnstorming days of aviation, the adventure of people flying as

an end in itself has captured the imagination of the enthusiasts and shaped the future they foresee. The Shuttle was designed in part to recreate the experience of flight for "everyman," either by allowing earthbound citizens to fly vicariously with the aviator or astronaut or to actually take them along for a joyride themselves.²⁰ It was for this that we have been subjected to the spectacle of the senator in space, the congressman in space, the Saudi prince in space, and finally the teacher in space. The tragedy of that last adventure spared us the journalist in space, proceeding inexorably to the man-in-the-street in space. About these adventures hung the same carnival atmosphere that greeted the barnstormers of the 1920s, who flew into Backwater township to take Aunt Hazel up for her first and usually last spin among the clouds. The real future of aviation, however, lay elsewhere. It was designed not at the air races or the fair ground but in the government research laboratories and the aircraft manufacturing plants. It was created not by barnstormers and recordbreakers -- though they played their part -- but by the engineers and businessmen and soldiers who transformed the airplane from a toy and a curiosity into a practical vehicle for transportation and national security.²¹ The "airframe revolution" of the early 1930s which gave us the DC-3 transformed the airplane from circus horse into workhorse.²² War and commerce shaped the future of aviation. So too will they shape the future of spaceflight.

The analogy between barnstorming and the romantic era of spaceflight should not be pushed too far, if only because spaceflight has not yet emerged from its romantic phase. Neither war nor commerce has yet provided the necessary spur to rational exploitation of space. Though Star Wars looms on the horizon, it is yet unclear when or if it will come to pass. Reconnaissance has historically been the most important military use of space,²³ and though this undertaking has expanded the military space budget to more than twice NASA's²⁴ it does not of itself promise to support the order-of-magnitude increase in activity that would be necessary to expand spaceflight in the way that military aviation grew in the middle of the twentieth century. On the civilian side, commercial activities seem to have reached a plateau beyond which they are not likely to advance in the current technological dispensation. Communications satellites remain an important commercial technology, but their future growth is more likely to be qualitative than quantitative. Failing a major national commitment to large-scale use of solar power collection in space, it seems unlikely that any commercial venture in space is going to greatly increase the development of "infrastructure" there.²⁵

Surely weather satellites and earth resources satellites will continue their incremental development, but neither seems likely to be a major commercial success. Talk of manufacturing in space, which has been going on for a quarter century or more, is still just that -- talk.

How, then, and when might the transformation to the second era of spaceflight be completed? Again, the answer is unclear. But it is apparent that the change is being delayed by our current national policies. The institutional home of our civilian space program is locked into a climate of opinion bred of Sputnik, Gagarin, and Apollo. It is intent upon extending the romantic era of spaceflight -- indeed upon building our whole future in space around a program of barnstorming. We are in a state of suspended adolescence, deferring mature exploitation of space in a childish infatuation with circus.

The origins of this anachronism are in the Apollo program. When Thomas O. Paine succeeded James E. Webb as administrator of NASA in 1969 he found the space agency in a paradoxical situation. Just months before the first moon landing, on the eve of one of the greatest technological triumphs of all time, NASA was experiencing budget cuts. In fact the agency's budget had been falling steadily since 1965, the midpoint of the Apollo undertaking.²⁶ Instead of the ambitious and exciting adventure in space that Apollo had seemed to promise, NASA faced a diminished and cloudy future. The second half of the 1960s had brought the United States a debilitating war abroad, domestic unrest, rioting in major cities, environmental crisis, and spectacular shortfalls in the federal budget. President Johnson had turned a deaf ear to proposals for an ambitious follow-on program to Apollo and Congress had called with increasing emphasis for practical payoffs on the \$25 billion dollars it had already invested in getting to the moon.²⁷ Congressman Joseph E. Karth, Chairman of the Space Science and Applications Subcommittee predicted in September 1969 that "the pressure on Congress to reduce the space budget still further will increase unless the orientation of NASA is based less on space spectacles and more on the production of tangible and economic benefits."²⁸ The success that awaited NASA in the summer of 1969 was to be rewarded with retrenchment.

Administrator Paine was not about to oversee the dismantling of the Apollo team. He was a space enthusiast of the first order, driven like many of his colleagues in NASA to pursue what they perceived as man's destiny in the heavens. If the country faced a succession of demoralizing crises in the late 1960s, then it was all

the more in need of Apollo, a spectacular national success in a decade gone sour. Richard Nixon, an old friend of the space program, was in the White House and no doubt open to a proposal that might move beyond the moon mission which was, after all, the creature of his former nemesis John Kennedy. In short, 1969 presented an opportunity for Paine and NASA to map out a new future for the agency, one more exciting than the politicians were able to see from the well of gloom into which they had fallen.

Paine quickly pressed President Nixon for a major new initiative in space. Nixon hesitated, appointing instead a Space Task Group to study the problem under the direction of his vice president, Spiro Agnew. In September 1969, two months after the first Apollo landing, the STG came in with its report.²⁹ It recommended Mars and a Space Station; the main purpose of the Space Station was to go to Mars. The report called for a balanced program that included space science and applications, but the overall pace and direction of the agency's agenda was tied to a barnstorming tour of the Red Planet.

The first choice of the Task Group was a commitment then and there to a manned Mars mission. Like Apollo, on which it was clearly modeled, this mission would send men to Mars as an end in itself.

Space science played a role, but the most important goal was to develop "new challenges for man in space."³⁰ (Read barrel rolls and Immelmann turns in the shadow of Phobos.) If a Mars landing in the early 1980s was too ambitious, the Task Group continued, the country could commit itself to a less expensive, more slowly paced program that would end up on Mars in the second half of the 1980s. Failing those two options, the President could simply opt for a Space Station and a Shuttle to ferry crew and supplies to and from the Station. The principal rationale for a Space Station since the nineteenth century had been as a way-Station to the planets, so even this third option supported the same goal.³¹

The Space Task Group echoed the NASA philosophy of the 1960s. It made manned spaceflight an end in itself and it proposed a series of "logical steps" as the centerpiece of the civilian space program in the decades to come. Beginning with Neil Armstrong's "one small step for [a] man, one giant leap for mankind," each successive step was a manned adventure, eschewing incremental advance in favor of quantum leaps in technological capability and developmental goals. Apollo had demonstrated that we could do anything we set our minds to. All

that was left was to lay out the agenda and get to work. The enterprise was simply too important and too exciting to be left to the timid and cautious.

The mentality that was driving this proposal appeared most clearly in a planning conference held the following June at the NASA research facility on Wallops Island, Virginia.³² There Paine called together his top executives for a brainstorming session on the future of "Spaceship NASA." Wernher von Braun lamented before his colleagues the "cruise ship model of human existence" then gripping the country, an apparent reference to building a launch vehicle like the Saturn and then using it for nothing more exciting than the Skylab Space Station. Paine proposed instead "fighting ships . . . both naval and buccaneering as the non-cruise ship nautical model for NASA."³³ He asked his colleagues to "consider NASA as 'Nelson's 'Band of Brothers' -- Sea Rovers' of 'swashbuckling buccaneer courage' who would 'stake out and create powerful outposts of stability, sanity and real future value for mankind in the new uncharted seas of space and global technology.'" Though the conferees identified some useful and practical technological objectives, such as new methods of food synthesis and improved engines and generators, it is nonetheless clear that they were in the throes of a romantic enthusiasm of Byronic proportions, a climate of opinion that left them deaf and blind to alternative futures in space. Men were going to sail on what President Kennedy called "this new sea;"³⁴ nothing else would do. Their vision of the future of spaceflight and how to achieve it made the modest goals of the barnstormers pale by comparison.³⁵

The wonder is not so much that NASA executives should think in such terms in 1970; in the intoxicating wake of the first moon landing even sober observers like Arthur Schlesinger, Jr., were given to hyperbole and flights of imagination. What is remarkable is that the agenda laid out in that first flush of enthusiasm has continued to hold sway over NASA through all the succeeding years. The agency is still committed to the succession of "next logical steps" envisioned in the Space Task Group report of 1969.³⁶ It settled for the Shuttle as the major undertaking of the 1970s, characterizing it as part of a space transportation system designed to support a Space Station, which was in turn a way Station to Mars. When the Shuttle became "fully operational" after its fourth flight in 1982, NASA immediately pressed the Reagan administration to authorize the "next logical step," the Station,³⁷ which the President approved in his State of the Union Address in 1984.³⁸ Now, with the Station hardly beyond the design stage,

NASA is already arguing about whether to go back to the moon next or proceed directly to Mars.³⁹

The problem with this agenda is that it is driven by romance not practicality. There are many worthwhile things to do in space; sending people there is one of the most expensive and least productive. It perhaps provides psychic income to those who thrill to such adventures, but the practical returns are far more difficult to identify. Like Charles Lindbergh's flight across the Atlantic or Amelia Earhart's failed attempt to circumnavigate the earth, these flights are primarily stunts to show that we can do them. After the moon landings, few people doubt that we are capable of sending men to Mars. If it were free, it would no doubt be worth doing. But in fact it and the other "logical steps" NASA has been pursuing for almost twenty years are extremely dangerous and ruinously expensive.

It costs ten times as much to conduct a space mission with people as it does with automated spacecraft.⁴⁰ The reason is simple. On a manned spaceflight you not only have to carry the extra weight of the crew but also the food, water, medicine, oxygen, and life-support equipment on which their survival depends. Additionally, and more importantly, you have to "man-rate" the spacecraft, i.e., build in redundant features, maintain rigid quality control on the manufacture of all components, and ensure reliability to levels that would be simply impractical on an unmanned vehicle. Just the documentation for a manned spaceflight is an administrative and financial burden that raises costs by an order of magnitude. George Low, the late head of manned spaceflight, Deputy Administrator of NASA, and stalwart defender of man in space once told a colleague that "it had cost 10 times as much to prepare a magnetometer for the Apollo project as it had to prepare a similar one for an unmanned project."⁴¹

Over the past quarter century, two-thirds of our space dollars have been invested in manned spaceflight, with little to show for the investment save circus.⁴² By and large it has been wonderful circus, just as the barnstorming was, but hardly more productive. The major achievement of the astronauts and cosmonauts to date is the accumulation of physiological data to determine their ability to survive long periods in space, should we ever find a reason for sending them there. The real payoff in space -- the work of the communications satellites, the weather and earth resources satellites, the scientific probes -- has been funded by the remaining one third of the civilian space budget.

NASA offers two good reasons for its continuing concentration

on manned spaceflight; behind them lurk the two real reasons. The first good reason is the man with a screwdriver. We need people on our spacecraft, so the argument goes, to respond to the unexpected. People are versatile and adaptive; machines are programmed and inflexible. Skylab is the ultimate example.

Launched unmanned on 14 May 1973, Skylab suffered serious damage on ascent and arrived in orbit with part of its meteoroid shield torn away, one of its two solar wings lost, and the other jammed shut.⁴³ Underpowered and overexposed to sunlight, the craft heated up to uninhabitable levels and threatened to preclude the manned visits that were to follow. The first visit, launched just eleven days later, turned into a rescue mission of high drama and heroic performance, both by the astronauts in orbit and the ground crews that designed and directed the rescue. The astronauts deployed the remaining solar wing and placed a jury-rigged cover over the exposed portion of the fuselage; internal temperatures fell and power rose to levels adequate for operation of the spacecraft and its equipment. A disastrous accident had been turned into an exciting salvage and a new legend had been added to the lore of manned spaceflight. The resourcefulness and versatility of the astronauts *in situ* had saved the mission and demonstrated the value of men in space.

The other good reason for men in space is that manned spaceflight has become a highly visible and widely appreciated measure of technological achievement in the Cold War race for prestige. The National Aeronautics and Space Act of 1958 specifically enjoins NASA to make the United States "a leader" in space activity; surely Congress wanted to race the Russians in space when it adopted that legislation.⁴⁴ This was the primary motive driving the Apollo program at the outset,⁴⁵ but not even that conclusive victory seems to have settled the issue. The Soviet Union continues to support a large, highly publicized manned program, capturing in the years since Skylab most of the records for achievement and endurance in orbit. Though no outsider can claim to discern with confidence Soviet motives, it seems clear that the Russians still believe that they reap a significant harvest of international prestige from their manned spaceflight accomplishments. Secretary General Gorbachev's invitation to President Reagan for the United States to join the Soviet Union in a manned mission to Mars is in keeping with the same spirit that moved the Russians into the Apollo-Soyuz Test Project of 1975, to pose as a technological equal of the United States by matching their manned spaceflight achievements and participating with them in joint ventures.⁴⁶ Why the U.S. feels compelled to

keep running the same race over is less clear, but we seem nonetheless committed to matching the Russians Space Station for Space Station indefinitely.

Both of these arguments are compelling; both are seriously flawed. First of all, the man with a screwdriver usually costs more than he is worth, in part because he isn't worth all that much. For every Skylab mission that can be cited, there are a dozen manned missions where the people on board did not begin to pay their way in any practical sense. Far from being an asset on most space missions, astronauts are a liability. Not only does the mission have to sustain the extra weight and bulk of them and their life support mechanisms, it also has to follow a flight plan dictated primarily by the safety of the crew. It cannot go too far from Earth, as the Voyager spacecraft have. It cannot travel in harm's way, as Pioneer 10 did when it passed through the asteroid belt between Mars and Jupiter or when it exposed itself to the full strength of Jupiter's radiation fields, 10,000 times more intense than the Earth's Van Allen belts. It cannot shut down its systems in flight to save power, as most of our unmanned spacecraft do on long missions. Nor can it completely change its mission profile as Voyager 2 did on the way to the outer planets.⁴⁷ Both Voyager spacecraft are likely to be still transmitting to us in the 21st century from nine billion miles away; manned spaceflights are pedestrian by comparison.

Nor is the Skylab mission all that persuasive as a model for the utility of manned spaceflight. After all, the crew was able to repair the craft only because it could design the rescue on earth and load up with the proper equipment before launch. An astronaut on the Martian surface with the wrong screwdriver would be just as useless as a computer in the same position. If the astronaut took all the right tools, then someone on earth had anticipated all the problems; in which case you don't need the astronaut in the first place. The real model for human versatility in space is the Apollo 13 mission, in which the flight directors in Houston designed and effected the rescue. The astronauts simply went along for the ride, though their presence necessitated the rescue in the first place. Had it been an automated flight, the spacecraft, which was no longer able to perform its mission, would simply have been abandoned. This is the way most of our missions run, with controllers on earth guiding and reprogramming our automated flights to adapt to new and unforeseen circumstances. The human element is important, but it doesn't have to be *in situ*.

The Viking missions to Mars, the most expensive space science

undertaking of the 1970s and perhaps the most technically sweet experiment in the history of spaceflight, revealed more clearly than any other enterprise the power of American space technology and the capabilities of remotely controlled vehicles. In 1975 the United States launched two spacecraft from earth on the 500-million-mile missions to Mars.⁴⁸ Viking 1 arrived on 20 July 1976 and began to survey the Martian surface. Finding the original landing site unsatisfactory, it searched about for two weeks. When it had identified a suitable site, it detached a lander which flew to the surface and soft-landed. Deploying an array of instruments and sensors, the lander began to survey the Martian environment. Most importantly, perhaps, it sampled the soil in its vicinity and tested the material in an on-board laboratory, attempting to answer the fundamental question of the mission: was there, or had there ever been, life on Mars? The results were finally inconclusive, but they gave little encouragement to those who believed that the planet is or was capable of supporting life.

Viking 2 arrived on 3 September 1976. It too orbited the planet and successfully dispatched its lander to the surface, setting down at another site believed conducive to life. The results of its tests were similarly indeterminate. The two failures together fed the growing pessimism about the possibility of life on Mars.

If these spacecraft did not finally resolve the issue of life on the Red Planet, they nonetheless provided a model for how to operate in space. The spacecraft, indeed the whole mission, were designed by men and women who programed the vehicles to do what they would do if they were on the Martian surface or in Martian orbit. They imagined in advance what life would be like if it existed on Mars and how you would test for traces of it, and they designed their experiments accordingly. Just as astronauts would take certain equipment and supplies if they were going to Mars themselves, based on their expectations of what they would find there, so too did the scientists and engineers on earth build into the Viking spacecraft the capabilities and characteristics they expected would be needed.

The mission flew two spacecraft to increase the probability that at least one would get through. The project team negotiated in advance all the trade-offs between the desires of the scientists to run experiments and the insistence of the engineers that the spacecraft function properly. The engineers of necessity won the close calls. As it happened both craft worked almost flawlessly and operated well beyond their rated capacities. Through ingenious reprogramming from earth, the expected mission life of the two orbiters was extended from ninety days to almost two

years for Orbiter 2 and over four years for Orbiter 1, which sent back data until it finally ran out of fuel on 6 August 1980. Lander 1 sent back data for over six years, until it reportedly ran out of fuel in 1983.⁴⁹ These spacecraft did everything that was expected of them and more. If they failed to find life on Mars it was because there is no life there to find or people here on earth asked the wrong questions or put them in the wrong place. It is difficult to see how humans on the expedition might have done more or better. In fact humans on earth controlled the spacecraft throughout their flights and directed their activities once they had landed.⁵⁰

The Viking mission to Mars cost about \$1 billion in 1975 dollars, a very large sum by space science standards but a pittance on the scale of manned spaceflight. The Apollo program had cost \$25 billion in 1960s dollars just to go to the moon. A manned Mars mission would have cost at least several times as much. It would have brought back some Martian samples for study here on earth, but it is possible to build machines to do that, as the Russians did with their Lunar moon vehicle in the 1960s. The Viking mission to Mars demonstrates with compelling clarity the most fundamental weakness of manned spaceflight: any specific mission we can identify to conduct in space we can build a machine to do. And we can do it more quickly, more safely, and at a fraction of the cost of sending people up to do it.

The prestige argument for manned spaceflight is less specific than the man-with-a-screwdriver, and for that reason less easy to pin down. There is no doubt that a certain cachet attaches to manned spaceflight and that Apollo returned to the U.S. just what it went after -- the international prestige of being the best in space. It is also clear that at least in the public eye, manned spaceflight is more important than any other space activity in establishing preeminence in space. It is not clear, however, that the country is willing to "pay any price" as President Kennedy put it in 1961 in order to match the Russians, especially now when peace is breaking out all over the world.

Furthermore, the emptiness of manned spaceflight is beginning to catch up with its reputation. Cosmonauts have been around the earth some 32,000 times now and they have virtually nothing to show for it save atrophied muscles and frazzled psyches. Their comings and goings to and from their Space Stations are reported in the press but given little attention. When the Challenger exploded in 1986, the major networks were not covering the launch because the ratings were terrible. Christa McAuliffe was needed on the flight in part to boost public interest in what was

becoming a ho-hum performance. The same public indifference even plagued the later Apollo missions, arguably the most exciting space ventures to date. Prestige may well attach to these undertakings, but it is the ability to do it, not the repetition that captures the public imagination. If we are going to have manned flights, we should reserve them for those unusual and demanding missions that might really benefit from human presence; leave the routine flights to the machines. This would entail no loss in international prestige; on the contrary, it might revive international interest. Prestige is an arguably rational goal for manned spaceflight; it is not a good reason for building our whole space program around manned spacecraft and facilities in space.

The real reasons why NASA has built the civilian space program around a succession of manned space spectacles are seldom discussed in public. First, it has been an article of faith within NASA since the budget cuts of the late 1960s that manned spaceflight sells. NASA and its supporters have been trying for twenty years to recapture the level of funding they enjoyed in the immediate aftermath of the Kennedy commitment to the moon mission. They view that as the appropriate level of national support for this crucial undertaking, and they view all budgets since as something of a cutback. The only way to return to those budget levels, they argue, is for the country to commit itself once more to a major, manned undertaking like Apollo. They made that argument when selling the Shuttle; they made it again in selling the Space Station; and they are making it now in selling a Mars mission as a follow-on to the Station.

This rationale is flawed on two levels. First, it doesn't work. Apollo was a crash program designed to address a unique set of historical circumstances. Five administrations and seven congresses have made it clear that the country is not likely to engage in such an undertaking any time soon, surely not on a regular basis. Rather, the country has supported NASA at a steady-state level. Since 1973 the civilian space budget has been approximately \$7.8 billion in constant 1988 dollars; it has not varied more than 10 percent above or below that figure, save for the extra funding provided in the aftermath of the Challenger accident.⁵¹ President Nixon's staff explicitly told NASA when it was negotiating for the Shuttle that it could expect a steady-state budget over the term of his presidency. Other administrations have tacitly told NASA the same thing, and Congress has gone along. Grandiose proposals like the early shuttle configurations and the early Space Station plans only get pared down to bargain basement versions then struggle along trying to reach completion

under the existing budget ceiling. The result is scaled-down, stretched-out, and unsatisfactory manned projects and the squeezing of other NASA programs to try to keep the spectacles alive. We watched this all through Shuttle development and that history is now repeating itself with the Station.

The rationale fails on yet another level because NASA has made of it a self-fulfilling prophecy. Believing that only manned spaceflight sells before Congress and the public, it has never given proper exposure to its unmanned programs. NASA's truly spectacular achievements in space science, earth resources monitoring, geodesy, and weather analysis have gone comparatively unnoticed. Public relations at NASA is built around astronauts, in part because the public shows the greatest interest in them, but in part also because NASA has always put them out front. If NASA invested as much effort in selling its unmanned programs, it would have a wider and more solid base of support than it does now.

But the final, real reason that NASA has built the civilian space program around a succession of manned space spectacles is a climate of opinion. The agency continues in the grip of an emotional enthusiasm for the spectacle of manned spaceflight itself. NASA is filled with space enthusiasts, sharing with Tom Paine an enthusiasm and zeal for peopling the universe that approaches a religious faith. Man's destiny is in the stars, they believe, and it is their mission to put him there. Accepting anything less would be an abomination of human nature and a compromise with destiny. They consider themselves to be, as Paine put it on Wallops Island, the keepers of the "boldest and most visionary goals," entrusted with "moving the planet into a better 21st century." They have no conception of why man should go into space, nor what he will do once there, but they are determined that he must go nonetheless.

This is the "winged gospel" of the barnstormers translated into the late twentieth century. In it are the seeds of our future. In it also is much silly fluff, romantic nonsense that will remain science fiction long after other, unimagined avenues in space are opened up and explored. Distinguishing between the fact and fiction entails hard calculations about cost effectiveness and the utility of what we are trying to accomplish in space. For only when the activity starts paying off on earth will it attract the funding necessary to realize its full potential. The issues of war and peace that are likely to provide that incentive are now being clouded by our romantic enthusiasms. It is time to grow up and get on with the business of space exploration and exploitation.

NOTES

1. Public Papers of the Presidents of the United States, Richard M. Nixon, 1969 (Washington, D.C.: General Services Administration, 1971), p. 542.

2. Carl Becker, *The Heavenly City of the Eighteenth-century Philosophes* (New London: Yale University Press, 1932), p. 5.

3. Henry Adams, *The Education of Henry Adams: An Autobiography* (New York: Houghton-Mifflin, 1988), p. 380.

4. Lewis Straus speech of 16 September 1954, quoted in Daniel Ford, *The Cult of the Atom: The Secret Papers of the Atomic Energy Commission* (New York: Simon and Schuster, 1982), p. 50.

5. Clifton R. Wharton, Jr., "The Green Revolution: Cornucopia or Pandora's Box?" *Foreign Affairs* 47, 1969: 464-76.

6. Military leaders have said the aerospace plane could reduce launch costs by 99%, the same kind of claim that was made for the shuttle. See Stephen W. Korthals-Altes, "Will the Aerospace Plane work?" *Technology Review* 43, January 1987.

7. The best survey of American aviation history is Roger Bilstein, *Flight in America, 1900-1983: From the Wrights to the Astronauts* (Baltimore: The Johns Hopkins University Press, 1984). Bilstein concentrates on the developments that were to produce real advances in aviation; consequently he devotes little attention to the barnstormers. Note, however, his treatment of "The Adventurers," pp. 83-85, and his observation that "in headlines and crowds, the men and women who flew the planes received the sort of acclaim granted to astronauts a generation later." A fuller treatment of the romantic era of aviation is C.R. Roseberry, *The Challenging Skies: The Colorful Story of Aviation's Most Exciting Years, 1919-1939* (Garden City, N.Y.: Doubleday, 1966), esp. chap. 4, "The Rollicking Barnstormers."

8. Charles H. Gibbs-Smith, *The Aeroplane: A Historical Survey of Its Origins and Development* (London: Science Museum, 1960), p. 59.

9. Chuck Yeager did not become a household word until Tom Wolfe popularized him as the ultimate holder of the right stuff. Tom Wolfe, *The Right Stuff* (New York: Farrar, Straus, Giroux, 1979).

10. For military aviation, Michael Sherry uses the term "the age of fantasy," similar in many ways to what I mean by the romantic era of spaceflight. See his *The Rise of American Air Power: The Creation of Armageddon* (New Haven: Yale University Press, 1987), chap. 1.

11. Joseph J. Corn, *The Winged Gospel: America's Romance with Aviation, 1900-1950* (New York: Oxford University Press, 1983).
12. *Ibid.*, p. 10.
13. *Ibid.*, p. 32.
14. *Ibid.*, pp. 32-33.
15. *Ibid.*, chap. 5.
16. *Ibid.*, chap. 6.
17. *Ibid.*, p. 32.
18. Charles D. Benson and William Barnaby Faherty, *Moonport: A History of Apollo Launch Facilities and Operations* (Washington, D.C.: NASA SP-4204, NASA, 1978), p. 529.
19. George E. Mueller, "The New Future for Manned Spacecraft Developments," *Astronautics and Aeronautics* 28, March 1969.
20. See Tom Wolfe, "Everyman vs. Astropower," *Newsweek*, February 10, 1986: 40.
21. Sherry, *The Rise of American Airpower*, chap. 1.
22. Bilstein, *Flight in America*, pp. 85-96. See also John B. Rae, *Climb to Greatness: The American Aircraft Industry, 1920-1960* (Cambridge: MIT Press, 1968).
23. In a famous gaffe in 1967, Lyndon Johnson revealed that the \$35-40 billion the United States had then spent on space was worth, in his opinion, ten times that much because the reconnaissance satellites had revealed information about Soviet military power that kept us from taking unnecessary military steps of our own. Chalmers M. Roberts, *The Nuclear Years* (New York: Praeger, 1971), p. 87.
24. The avalanche of funding for Apollo that began in fiscal year 1961 raised NASA's budget above the level of military spending on space in that year. NASA's budget remained above the military space budget until fiscal year 1982; four years later it was twice NASA's. *Aeronautics and Space Report of the President, 1987*, appendix E.
25. On the concept of infrastructure in space, see Office of Technology Assessment, *Civilian Space Stations and the U.S. Future in Space* (Washington, D.C.: OTA, 1984).
26. It was \$5,137.6 million in 1965, \$3,822 million in 1969. This represented a drop of 26% in actual dollars and 34% in constant dollars. *Ibid.*
27. W. Henry Lambright, *Presidential Management of Science and Technology: The Johnson Presidency* (Austin: University of Texas Press, 1985), pp. 141-51.

28. Quoted in Ken Hechler, *The Endless Space Frontier: A History of the House Committee on Science and Astronautics, 1959-1978*, abr. by Albert E. Eastman, AAS History Series, Vol. 4 (San Diego, CA: American Astronautical Society, 1982), p. 293.
29. *The Post-Apollo Program: Directions for the Future, Space Task Group Report to the President* (Washington, September 1969).
30. *Ibid.*, p. 5.
31. Barton D. Hacker, "And Rest As on a Natural Station: From Space Station to Orbital Operations in Space Travel Thought, 1895-1951," unpublished paper [1972], NASA History Office Archives, Washington, D.C.
32. Memorandum, NASA Deputy Associate Administrator to Program Associate Administrators, on "Principal Wallops Island Action Items," undated, in von Braun Library/Archive, A-SRC, Huntsville, AL. The file folder indicates that the meeting was 11-14 June 1970.
33. Paine is a student of naval history, with a large personal library and at least one publication in the field: *Submarining: Three Thousand Books and Articles* (Santa Barbara, CA: General Electric Co.-TEMPO, Center for Advanced Studies, [1971]).
34. *Public Papers of the Presidents of the United States, John F. Kennedy, 1962* (Washington, D.C.: General Services Administration, 1963), p. 669.
35. The swashbuckling episode was first made public by Homer Newell in his *Beyond the Atmosphere: Early Years of Space Science* (Washington, D.C.: NASA SP-4211; NASA, 1980), p. 288. Newell notes that Paine later questioned the appropriateness of the term.
36. "The Next Logical Step," NASA Brochure (Washington, D.C.: NASA, no date)
37. James M. Beggs to Edwin Meese, III, May 21, 1982; copy in the possession of the author.
38. "The State of the Union," January 25, 1984, *Weekly Compilation of Presidential Documents* 20, (January 30, 1984): 84.
39. See, for example, the Ride report and the swirl of controversy that surrounded it. *Leadership and America's Future in Space, A Report to the Administrator* (Washington, D.C.: NASA, August 1987).
40. Newell, *Beyond the Atmosphere*, pp. 162-63.
41. *Ibid.*, p. 390.
42. Hans Mark and Arnold Levine, *The Management of Research Institutions: A Look at Government Laboratories* (Washington, D.C.: NASA SP-481, NASA, 1984). One recent researcher put the figure even higher, at 80%; see T.A. Heppenheimer, "Son of Space Station," *Discover*, (July 1988): p. 64.

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43. W. David Compton and Charles D. Benson, *Living and Working in Space: A History of Skylab* (Washington, D.C.: NASA SP-4208, NASA, 1983).

44. Robert L. Rosholt, *An Administrative History of NASA, 1958-1963* (NASA SP-4101, Washington, D.C.: NASA, 1966); Allison Griffith, *The National Aeronautics and Space Act: A Study of the Development of Public Policy* (Washington, D.C.: Public Affairs Press, 1969).

45. Vernon Van Dyke, *Pride and Power: The Rationale of the Space Program* (Urbana: University of Illinois Press, 1964); and John Logsdon, *The Decision To Go To The Moon: Project Apollo and the National Interest* (Chicago: University of Chicago Press, [1970] 1976).

46. Edward Clinton Ezell and Linda Neuman Ezell, *The Partnership: A History of the Apollo-Soyuz Test Project* (NASA SP-4209, Washington, D.C.: NASA, 1978); Walter A. McDougall, *... the Heavens and the Earth: A Political History of the Space Age* (New York: Basic Books, 1985).

47. Michael Gold, "Voyager to the Seventh Planet," *Science* 86 (May 1986): 32-38.

48. Edward Clinton Ezell and Linda Neuman Ezell, *On Mars: Exploration of the Red Planet, 1958-1978* (NASA SP-4212, Washington, D.C.: NASA, 1984).

49. Edward Hutchings, Jr., "The Autonomous Viking," *Science* 219 (February 18, 1983): 803-808; *New York Times*, May 6, 1983, p. B6; *ibid.*, 23 May 1983, p. A17.

50. The story circulates in the aerospace community that the Viking 1 lander did not really run out of fuel. Rather an errant signal from a ground controller on Earth pointed the antenna toward the Martian surface, cutting off forever its ability to communicate. If true, this means it was human error that terminated this sweetest of automated missions. Of course, an astronaut on Mars with a screwdriver could have righted the antenna, but for the cost of sending and maintaining him there for six years we could send dozens of replacement landers.

51. It is interesting to note, however, that the additional funding received in the wake of the Challenger accident to fix the shuttle program seems to have become permanent, amounting to something like a 25% real increase in the NASA budget over two years in the era of Gramm-Rudman.

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